



The Experimental Role of Medicinal Plants in Treatment of *Toxoplasma gondii* Infection: A Systematic Review

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Abstract

Background *Toxoplasma gondii* is the global protozoa that could cause contamination in warm-blooded animals and is considered among the opportunistic pathogens in immunocompromised patients. Among the people at risk, toxoplasmosis infection can lead to the incidence of severe clinical manifestations, encephalitis, chorioretinitis, and even death.

Purpose The present research is focused on the new research for the treatment of toxoplasmosis parasitic disease using medicinal herbs.

Methods The search was performed in five English databases, including Scopus, PubMed, Web of Science, EMBASE, and Google Scholar up from 2010 to December 2019. Studies in any language were entered in the searching step if they had an English abstract. The words and terms were used as a syntax with specific tags of each database.

Results Out of 1832 studies, 36 were eligible to be reviewed. The findings showed that 17 studies (47%) were performed *in vitro*, 14 studies (39%) *in vivo*, and 5 studies (14%) both *in vivo* and *in vitro*.

Conclusion The studies showed that the plant extracts can be a good alternative in reducing the toxoplasmosis effects in the host and the herbal extracts can be used to produce natural product-based drugs affecting toxoplasmosis with fewer side-effects than synthetic drugs.

Keywords *Toxoplasma gondii* · Medicinal plants · Systematic review · Herbal medicine · Treatment

Introduction

Toxoplasma gondii is the compulsory and intracellular protozoa in all nuclear cells of the vertebrates, including humans and mammals [1]. Two forms of these protozoa are clinically found in human beings, tachyzoite (infectious, rapid proliferation) [2] and bradyzoite (dormant, found in

tissue cysts) [3]. Although most of the toxoplasmosis cases in healthy people are asymptomatic or mild asymptomatic, in those first involved with parasites, such as pregnant women, the infants received damage, such as microcephaly, hydrocephalus, blindness, abortion and death [4]. Although pyrimethamine and sulfadiazine can be used in the treatment of toxoplasmosis caused by this parasite, due to the side effects of these drugs, especially during pregnancy, they are limited [5]. However, the disadvantages of these drugs are related to the limited effectiveness in parasite purification and low penetration of spiramycin in embryonic tissues as well as poisoning caused by pyrimethamine and sulfadiazine [6]. Numerous studies have been performed based on the use of medicinal herbs for the treatment of toxoplasmosis, but more extensive research seems to be needed to investigate for the effective substances and toxic substances found in medicinal herbs [7].

Thus far, the extracts of various medicinal herbs, such as *Artemisia annua* [8], *Zingiber officinale* [7], *Sophora flavescens* Aiton [9] and *Eurycoma longifolia* [10], have been

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investigated and it has been shown that they are effective against the RH strain of *Toxoplasma gondii*. Albeit many studies have been done for the vaccine, but, no vaccine has been already available to prevent the spread of disease and it is not appropriate treatment for eradicating chronic infection among the infected individuals [11]. Therefore, the new drugs are required to treat *T. gondii* infection. Nowadays, some advances can be found in the use of medicinal herbs in the treatment of infectious diseases, especially in traditional Chinese medicine [9]. Hence, studies can promise the development of new drugs based on medicinal herbs in the treatment of infectious diseases. This paper attempts to describe the latest research carried out on *T. gondii* infection and new therapy methods using medicinal herbs and the mechanism of the effect of drug extracts *in vivo* and *in vitro* on the *T. gondii* strains. Besides, due to the presence of people with immunodeficiency, pregnant mothers and organ transplant patients who are susceptible to *T. gondii*, it seems necessary for future works use medicinal herbs in preventing and developing new therapeutic strategies; the present research is focused on the new research for the treatment of toxoplasmosis parasitic disease using medicinal herbs.

Materials and Methods

Search Strategy

The current study was carried out 06-PRISMA guideline (Moher *et al.* 2009) and registered in the CAMARADES-NC3Rs preclinical systematic review and meta-analysis facility (SyRF) database. The search was performed in five English databases, including Scopus, PubMed, Web of Science, EMBASE, and Google Scholar up from 2010 to December 2019. The words and terms were used as a syntax with specific tags of each database. The searched words and terms were: “*Toxoplasma gondii*”, “Medicinal Plants”, “Preparation”, “Stem bark”, “Concentration”, “*in vitro*” and “*in vivo*”.

Studies Selection

Those studies were examined in which the effect of Medicinal plants was measured on “*Toxoplasma gondii*”. First, the studies were imported to the EndNote X7 software (Thomson Reuters, New York, NY, USA) and duplicate studies were deleted. Afterwards, three independent authors examined the title and abstract of the studies and the relevant studies were included for further analysis. The same authors carefully read the studies and the eligible studies with adequate inclusion criteria were selected. Corresponding author resolved any disagreement between the authors.

Inclusion Criteria

Studies in any language were entered in the search step if they had an English abstract. Inclusion criteria in this investigation were the *in vitro* and *in vivo* studies evaluating the effects of herbal medicines on toxoplasmosis.

Exclusion Criteria

The studies with poor methodology, inadequate data, inappropriate analysis, only abstract, failure to match methods with results, the incorrect interpretation of the results, and too much emphasis on the results were excluded from the study (Fig. 1).

Data Extraction

Three independent authors extracted information from the selected articles and, if needed, the differences were resolved by the corresponding author. The extracted data include: botanical name, preparation, organism (strain) tested, part of used, concentration, exposure time, positive control dose, type of animal, concentration, treatment durations, routes of administration, infection routes (cysts or tachyzoites), and results.

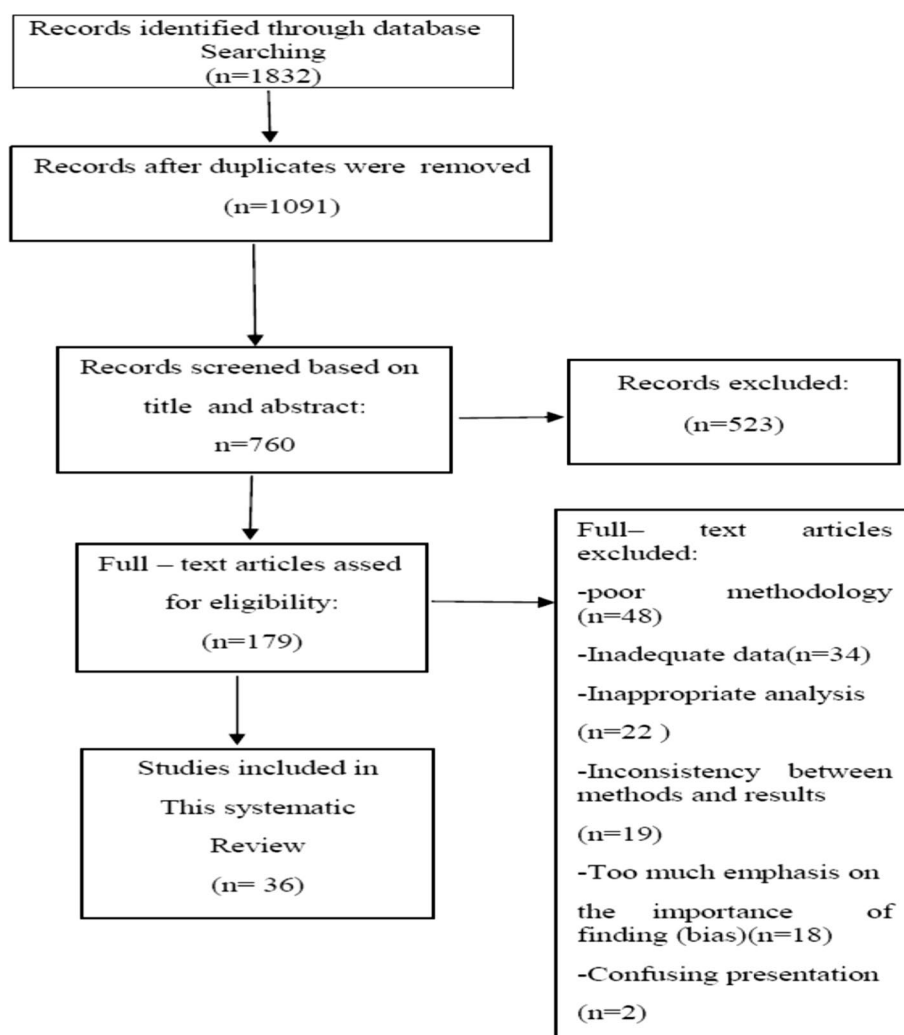
Results

Of the 1832 papers obtained from all of the databases between 2010 and 2019, 36 articles were reliable and were scrutinized (Fig. 1). Of these, 17 (47%) *in vitro* and 14 (39%) *in vivo* and 5 (14%) *in vitro* and *in vivo*. *In vitro* efficacy of herbal drugs against *T. gondii* was evaluated in Table 1, whereas *in vivo* efficacy of medicinal herbs tested against *T. gondii* shown in Table 2.

From 17 *in vitro* studies, the most used preparation method was ethanolic method (7 studies = 41%), and methanolic preparation with 6 studies (35%) has the highest issue, whereas among 14 *in vivo* studies, methanolic preparation with 4 (29%) studies and essential oil with 3 studies (22%) were the most applied methods.

Of the 19 *in vivo* studies, only 2 studies performed with rats, the other experiments were using different mice strains, such as ICR, NMRI, Balb/c, Swiss albino, etc. Among them, most are studies of the ICR (3 = 16%), as well as NMRI (3 = 16%) race. Most of the studies were conducted on leaves [10 studies], followed by roots [9 studies]. More details have been presented in Tables 1 and 2.

Fig. 1 A flowchart depicts the study design process. The PRISMA flowchart for searching, screening, and selecting the studies



Discussion

The current therapeutic study research on various strains of *T. gondii* using plant extracts derived from local herbs of different parts of the world. The *Toxoplasma* parasite is resistant to a wide range of inhibitors [44, 45], and long-term use of drugs causes drug resistance in parasites, including *T. gondii* [46]. Moreover, the high cost of chemical drugs as well as less access to these drugs, especially in poor countries, the use of local herbs for therapeutic purposes is increasing [47]. Medicinal herbs are the source of new drugs, which might increase the normal antimicrobial effects, largely reduce the costs, improve treatment quality and decrease drug side effects. Meanwhile, the pharmaceutical industry is looking for alternative drugs to deal with parasitic diseases or to improve the natural product-based drugs against parasitic diseases. In recent years, various medicinal herbs have been investigated to detect their potential anti-parasitic properties [48]. The continuous efforts for the exploitation of current medicinal

herbs, searching for new medicinal herbs and further production and evaluation of the anti-toxoplasma effects of these drugs can be more accessible alternatives for the treatment with synthetic standard drugs.

Securinega suffruticosa

This plant is used in traditional Chinese medicine combined with other herbs, which has been used for the treatment of diverse diseases, including rheumatism and inflammation caused by infectious diseases [49]. Securinine also has an inhibitory effect against *P. falciparum* [50]. In the study by Holmes *et al.*, the role of securinine alkoid products derived from *S. suffruticosa* against *T. gondii* was noted. In this research, the anti-inhibitory effects of Securinine toxoplasmosis *in vitro* indicated the inhibitory effects of toxoplasmosis in comparison with pyrimetamine, and also transforming tachyzoites forms into bradyzoites [12].

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Sophora flavescens</i> oxymatrine (OM) matrine (ME)	Alkaloids Alkaloids	RH RH	Pure pure	0, 10, 100, 200, 500 and 1000 µM 0, 10, 100, 200, 500 and 1000 µM	24 h 24 h	Spiramycin	IC ₅₀ = 843.44 µM IC ₅₀ = 1014.01 µM ME showed high anti- <i>T. gondii</i> activity IC ₅₀ = 936.31 µM	[9]
<i>Sorghum bicolor</i>	Methanolic	RH (YFP)	Leaves	3, 500 parasites/ml	48 h		For the crude extract (IC ₅₀ = 3.65 µg/ml), the hexane-methanol fraction (IC ₅₀ = 2.74 µg/ml), and the hexane fraction (IC ₅₀ = 3.55 µg/ml)	[14]
<i>Sambucus nigra</i>	Methanolic	RH	Fruits and leaves	5, 10, 25 and 50 mg/ml	30, 60, 120 and 180 min	Pyrimethamine	Fruits of <i>S. nigra</i> on mortality of <i>T. gondii</i> tachyzoites <i>in vitro</i> in 60 min at concentration 5 mg/ml = 90.58 ± 3.46 in 30 min at 10 mg/ml = 67.12 ± 1.54 leaves of <i>S. nigra</i> on mortality of <i>T. gondii</i> tachyzoites <i>in vitro</i> in 60 min at concentration 5 mg/ml = 73.37 ± 0.12 in 30 min at 10 mg/ml = 80.87 ± 1.81 In 30 min at 100 mg/ml = 98.64 ± 1096	[15]
<i>Myristica fragrans</i> Houtt	Essential oils	?	Seed kernel	3.125–100 µg/ml	24 h	Clindamycin	EC ₅₀ = 24.45 µg/ml EC ₅₀ = 16.57 µg/ml	[16]
<i>Myrtus communis</i> <i>Artemisia aucheri</i> Boiss	Ethanol Ethanol	RH RH	leaves aerial parts	10, 50, and 100 mg/ml 10, 50, and 100 mg/ml	10, 30, and 45 min 10, 30, and 45 min	pyrimethamine	Inhibitory effects against tachyzoites forms EC ₅₀ = 122.0 µg/ml Inhibitory effects against tachyzoites forms EC ₅₀ = 131 µg/ml EC ₅₀ = 0.176 µg/ml	[17]

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Eurycoma longifolia</i> TAF 355	Methanol	RH	Roots	0–100 µg/ml	24 h	Clindamycin 10.5 µg/ml	<i>In vitro</i> anti- <i>T. gondii</i> RH strain activity and selectivity of <i>E. longifolia</i> fractions	[10]
<i>TAF 401</i>	Methanol	RH	Roots	1.56–100 µg/ml	24 h		EC50 = 0.369 µg/ml EC50 = 0.882 µg/ml EC50 = 0.016 µg/ml	
<i>Achillea millefolium</i> <i>Hypericum perforatum</i>	Ethanol Ethanol	RH RH	Aerial parts Aerial parts	50, 100, or 200 µg/ml 50, 100, or 200 µg/ml	24 h 24 h	Pyrimethamine	<i>In vitro</i> anti-Toxoplasma activity EC50 = 215 µg/ml <i>In vitro</i> anti-Toxoplasma activity EC50 = 153 µg/ml EC50 = 0.176 µg/ml	[18]
<i>Ginkgo biloba</i> sars-cotesta	Ginkgolic acids	RH	Leave and s	3.13–100 µg/ml	24 h or 48 h		When 100 µg/ml GAS was added, the number of trophozoites decreased sharply. After 4 h, <i>Toxoplasma gondii</i> was found only rarely in the HFF cells	[19]

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Arecae pericarpium</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h	Sulfadiazine	<i>In vitro</i> anti- <i>Toxoplasma gondii</i> RH	[20]
<i>Sophora flavescens</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h	Pyrimethamine	EC50 = 1.30 mg/ml	
<i>Aiton</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h	Spiramycin	EC50 = 0.20 mg/ml	
<i>Melae cortex</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 0.77 mg/ml	
<i>Rheum undulatum</i> L.	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 2.28 mg/ml	
<i>Pulsatilla radix</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 1.52 mg/ml	
<i>Acorus gramineus</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 0.11 mg/ml	
<i>Solan</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 0.15 mg/ml	
<i>Dryopteris cras-sirizoma</i>	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 1.78 mg/ml	
<i>Quisqualis indica</i> L.	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 1.57 mg/ml	
<i>Fraxinus rhyncho-phylla</i> Hance	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 0.13 mg/ml	
<i>Glycyrrhiza glabra</i> L.	Methanolic	RH	?	0.11–2.28 mg/ml	24 h		EC50 = 1.53 mg/ml	
<i>Portulaca oleracea</i> L.							EC50 = 1.57 mg/ml	
<i>Schizandra chinensis</i> Baillon							EC50 = 0.18 mg/ml	
<i>Zingiber officinale</i>							EC50 = 1.40 mg/ml	
<i>Angelica gigas</i>							EC50 = 1.53 mg/ml	
<i>Artemisia scoparia</i>							EC50 = 0.07 mg/ml	
<i>Zingiber officinale</i> (GE)	Methanol	RH	Roots	30–240 µg/ml	24 h	Sulfadiazine	EC50 = 0.10 mg/ml	
<i>Zingiber officinale</i> (GE/F1)	Methanol	RH	Roots	30–240 µg/ml	24 h		EC50 = 0.31 mg/ml	
							Inhibitory concentration against RH	[7]
							IC50 = 220.83 µg/ml	
							Inhibitory concentration against RH	
							IC50 = 205.56 µg/ml	
							GE/F1 was more effective than other extracts	
							IC50 = 276.81 µg/ml,	
<i>Artemisia annua</i>	Artesunate	RH	?	0.1, 0.5, 1 and 2 µg/ml	24, 48 and 72 h		The percentage of viable tachyzoites of 0.1 µg/ml artesunate, after 24 h was 41.67%. That decreased after 48 h to 16.67% which was statistically significant ($p < 0.05$)	[8]

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Fraxinus rhycho-phylla</i>	Oleuropein	RH	?	0–1000 µg/ml	24 h	Sulfadiazine Pyrimethamine	<i>In vitro</i> anti-Toxoplasma gondii RH EC50 = 139 µg/ml EC50 = 600 µg/ml EC50 = 135 µg/ml	[21]
<i>Piper betle</i> L	Ethanol	RH-GFP and PLK-GFP	Leaves	1, 5, 10, 25, 50, and 100 µg/ml	72 h	Sulfadiazine	Anti-Toxoplasma activity of <i>P. betle</i> extract on RH-GFP and PLK-GFP respectively; IC50 = 23.2 g/ml IC50 = 21.4 µg/ml	[22]
<i>P. sarmentosum</i> Roxb	Ethanol	RH-GFP and PLK-GFP	Leaves	1, 5, 10, 25, 50, and 100 µg/ml	72 h		Inhibition of the parasite infection in <i>P. sarmentosum</i> = 94.7 ± 5.3%	
<i>Piper nigrum</i> L	Ethanol	RH-GFP and PLK-GFP	Seeds	1, 5, 10, 25, 50, and 100 µg/ml	72 h		Inhibition of the parasite infection in <i>P. nigrum</i> = 63.2 ± 5.3%	
							Anti-Toxoplasma activity of sulfadiazine extract on RH-GFP and PLK-GFP respectively IC50 = 99.4 µg/ml IC50 = 22.3 µg/ml	

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Psidium guajava</i> L	Alkaloids	RH	Leaves	200, 100, 50, 25, 12.5, 6.25, 3.125 µg/ml	24 h	Clindamycin	<i>In vitro</i> anti- <i>Toxoplasma gondii</i> RH	[23]
<i>Tinospora crispa</i>	Alkaloids	RH	Stem		24 h		EC50 = 121.18 ± 6.56 µg/ml	
<i>T. crispa</i> F1	Methanolic	RH	Stem		24 h		EC50 = 7.71 ± 1.56 µg/ml	
<i>T. crispa</i> F2	Methanolic	RH	Stem	200, 100, 50, 25, 12.5, 6.25, 3.125 µg/ml	24 h		EC50 = 34.24 ± 4.12 µg/ml	
<i>T. crispa</i> F3	Methanolic	RH	stem		24 h		EC50 = 22.98 ± 2.66 µg/ml	
<i>T. crispa</i> F4	Methanolic	RH	Stem	?	24 h		EC50 = 11.28 ± 1.62 µg/ml	
<i>T. crispa</i> F5	Methanolic	RH	Stem	?	24 h		EC50 = 6.71 ± 1.95 µg/ml	
<i>T. crispa</i> F6	Methanolic	RH	Stem	?	24 h		EC50 = 6.06 ± 2.51 µg/ml	
<i>T. crispa</i> F7	Methanolic	RH	Stem	?	24 h		EC50 = 7.34 ± 1.17	
<i>T. crispa</i> F8	Methanolic	RH	Stem	?	24 h		EC50 = 25.97 ± 0.28	
							Among the fractions that were tested for antiparasitic assay, F4 (EC50 = 6.71 ± 1.95 µg/ml) and F5 (EC50 = 6.06 ± 2.51 µg/ml) showed the most significant inhibitory effect on parasite growth	
<i>Tinospora crispa</i>	Ethanollic	RH	Stem	1.56–200 µg/ml	24 h	Clindamycin	EC50 = 6.24 ± 0.53 µg/ml The EETC had no cytotoxic effects on Vero cells with IC50 = 179 µg/ml, as compared to clindamycin (IC50 = 116.5 µg/ml) and <i>Veratrine alkaloid</i> (IC50 = 60.4 µg/ml) IC50 = 6.31 µg/ml in comparison with clindamycin (IC50 = 8.33 µg/ml) and <i>Veratrine alkaloid</i> (IC50 = 14.25 µg/ml)	[24]

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Tectona grandis</i> Tg-BE; <i>T. grandis</i> Tg-BH Tg-LH Tg-LE <i>Vernonia amygdalina</i> Va-LE	Ethanollic and hydro-Ethanollic Ethanollic Hydro-ethanollic Ethanollic Hydro-ethanollic Ethanollic	RH-b-gal RH-b-gal RH-b-gal RH-b-gal RH-b-gal RH-b-gal	Leaves and barks Bark Bark Leaf Leaf Leaf	1–200 µg/ml 1–200 µg/ml 1–200 µg/ml 1–200 µg/ml 1–200 µg/ml 1–200 µg/ml	24 h 24 h 24 h 24 h 24 h 24 h		Anti-Toxoplasma efficacy of the plant extracts IC50 = 15.3 ± 0.6 µg/ml IC50 = 176.7 ± 76.7 µg/ml IC50 = 143.3 ± 35.4 µg/ml IC50 = 59.8 ± 0.9 µg/ml IC50 = 5.6 ± 0.3 µg/ml Ethanollic extract of the leaves of <i>V. amygdalina</i> (Va-LE) was the most effective against <i>T. gondii</i> with an IC50 of 5.6 ± 0.3 µg/ml FII, at 0.3 mg protein/ml, was assayed it significantly ($p \leq 0.001$) reduced both the number of <i>T. gondii</i> infected cells and the number of intracellular parasites [25]	
<i>Jatropha curcas</i>	Oil	RH	Seed	0.01–3 mg/ml	24 h		Neem progressively reduced the infection by 2 mg/ml (60%), 3 mg/ml (75%) and 5 mg/ml (85%), respectively This infection reduction (70%) was similar to that seen for the 1 mg/ml concentration. At higher doses, a reduction of about 90% was observed [26]	
<i>Azadirachta indica</i> A. Juss. (neem) <i>Melia azedarach</i> L.	Distilled water Distilled water	RH RH	Leaves leaves	0.15, 0.3, 0.5, 1, 2, 3 and 5 mg ml 0.15, 0.3, 0.5, 1, 2, 3 and 5 mg ml	24 h 24 h		Neem progressively reduced the infection by 2 mg/ml (60%), 3 mg/ml (75%) and 5 mg/ml (85%), respectively This infection reduction (70%) was similar to that seen for the 1 mg/ml concentration. At higher doses, a reduction of about 90% was observed [27]	
<i>Astragalus</i> <i>Membranaceus</i> <i>Scutellaria baicalensis</i> GEORGI	Distilled water Distilled water	RH RH	Roots Roots	1, 10, and 10 mg/ml 1, 10, and 10 mg/ml	72, 96, and 120 h 72, 96, and 120 h	Cotrimoxazoleor SMX	AmE or SbE had significantly decreased intracellular replication at 72, 96, and 120 h after treatment ($p < 0.01$) SMX treated groups had significant lower living parasite numbers at all the times ($p < 0.01$) [28]	

Table 1 (continued)

Botanical name	Preparation	Organism (strain) tested	Part of used	Concentration	Exposure time	Positive control dose	Result	References
<i>Torilis japonica</i>	Ethanol	?	?	2.850 1.425 0.713	?	Sulfadiazine	<i>T. japonica</i> (TJ2) inhibited <i>T. gondii</i> proliferation by 99.2, 94.4, 88.6 and 27.0%. Four fractions of <i>S. flavescens</i> (SF1–SF4) inhibited <i>T. gondii</i> proliferation by 99.6–60.6, 96.9–48.1, 92.3–68.2 and 95.4–52.9% ?	[29]
<i>Sophora flavescens</i>	Ethanol	?	?	0.356 ng/ml 2.850 1.425 0.713 0.356 ng/ml				

Panax ginseng

Ginseng derived from *Panax ginseng* C.A. Meyer has been used for a long time as a traditional drug [51]. It is believed that Ginseng saponins, like ginsenosides, are one of the active parts of the root [52]. Ginsenosides also increase the response of the antibodies against viral and bacterial antigens [53].

In the study by Qu *et al.* [30], ginsenoside Re derived from *P. ginseng* along with recombinant ROP 18 proteins increased the production of IgG antibodies (IgG1 and IgG2a) as well as inflammatory cytokine IFN- γ in mice. This finding explains, the recombinant ROP 18 protein with ginsenoside Re caused a robust humoral and cellular immune response; whereas, the survival rate of this group of mice was higher than other groups [30]. Qu [32] used ginsenoside Rg1 to show that the mice immunized with recombinant surface antigens of *T. gondii* (rSAG1) in combination with Rg1 at the concentrations of 100 $\mu\text{g/ml}$ and 50 $\mu\text{g/ml}$ had longer survival time. Besides, in the group treated with (rSAG1/Rg1) at the concentration of 100 $\mu\text{g/ml}$, the cytokine level of IFN- γ (473.0 ± 11) was higher than other groups, which could be due to the ability of ginsenoside Rg1 as a promising vaccine against toxoplasmosis [32].

Zingiber officinale Roscoe

Zingiber officinale Roscoe, which is commonly called ginger, is used as a spice for cooking and treatment of common cold, coughs, headaches, diarrhea and digestive problems in Asian countries. The antioxidant, anti-inflammatory and anti-bacterial effects of this plant have been proven [7]. Published study by Choi *et al.*, the role of ginger extract (GE) and ginger-derived fraction GE/F1 against *T. gondii* both *in vitro* and *in vivo* showed GE/F1 strongly inhibited the proliferation of *T. gondii* in C6 cells and also reduced inflammatory cytokines *in vivo* [7].

Sorghum bicolor

Sorghum is the fifth most important cereal in the world and often grows in tropical regions, such as Asia, Africa and South America. It also contains an abundance of 3-deoxyanthocyanidins compounds [13]. Abugri declared, the rich extract of 3-deoxyanthocyanidins from *S. bicolor* red leaves *in vitro* had inhibitory and proliferative capability of *T. gondii* [13]. Abugri *et al.*, in another study, displayed that the *S. bicolor* oil-like crude extracts and fractions have potency against the tachyzoite stage of *T. gondii*, also, their results showed high antiparasitic effects of *S. bicolor* extracts with low cytotoxic effects [14].

Table 2 Included publications of survey on the efficacy and activity of herbal medicines used against Toxoplasmosis *in vivo*

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Panax ginseng</i> <i>C.A</i>	Physiological saline	RH	Root	Mice (ICR)	10, 50 and 100 μ g	15 days	i.p	Tachyzoite	The results showed that the immunogenicity and protection potency of recombinant protein ROP18 with the assistance of ginsenoside Re were comprehensively evaluated	[30]
<i>Sophora flavescens</i>	Alkaloids	RH	Aerial parts	Mice (KM)	100 mg/kg	4 days	i.p	Tachyzoite	After treated with 100 mg/kg of OM and ME, the numbers of tachyzoites in the peritoneal cavity of mice were significantly decreased	[9]
<i>Lycopodium clavatum</i>	Alcohol	ME49	Unknown	Mice (Mus musculus)	1 ml	3–4 days	Orally	Cysts	<i>L.clavatum</i> 200dH treatment in the G48 group was beneficial and provided a better clinical prognosis but resulted in a higher number of brain cysts at 60 dpi	[31]

Table 2 (continued)

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Ginsenoside</i>	?	RH	Root	Mice (ICR)	10,50 and 100 µg	2 Dose as vaccine	Subcutaneously	Tachyzoite	The results showed that the groups immunized with rSAG1 and Rg1 [50 g, 100 g] developed a high level of specific antibody responses against T. gondii rSAG1, a strong lymphoproliferative response, and significant levels of cytokine production	[32]
<i>Berberis vulgaris</i>	Methanolic	RH	Root	Mice (NMR1)	1 and 2 g/kg	14 days	Orally	Tachyzoite	The mortality rate of infected mice was 100%, in 9th day after orally administration of B. vulgaris at the concentrations of 1 and 2 g/kg	[33]

Table 2 (continued)

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Bunium persicum</i>	Essential oil	RH	Aerial parts	Mice (NMRI)	0.05 and 0.1 ml/kg	14 days	i.p	Tachyzoite	The difference in mean number of tachyzoites between infected mice received B. persicum essential oil and the control group was statistically significant ($p < 0.05$)	[34]
<i>Satureja khuzestanica</i>	Essential oil	RH	Aerial parts	Mice (NMRI)	0.2 and 0.3 ml/kg	2 weeks	Orally	Tachyzoite	SKEO significantly reduced the mean number of tachyzoites compared with control group ($p < 0.05$)	[35]
<i>Nigella sativa</i>	Black Seed oil	Me49	Seed	Mice (Swiss albino)	5 ml/kg	2 weeks before and day 4 post infection	Orally	Cyst	BSO prophylactic or therapeutic groups showed milder meningitis, encephalitis and perivascular cuffing compared to the infected untreated control ($p < 0.05$)	[36]

Table 2 (continued)

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Astragalus membranaceus</i> and <i>Scutellaria baicalensis</i>	Aqueous	RH	Root	Mice (ICR)	75 mg/kg and 50 mg/kg	21 days [2 dose]	i.p	Tachyzoite	Infected mice treated with AmE or SbE, or vaccinated mice and then challenged, had significantly prolonged survival time, decreased parasite burden, improved liver histopathological score, and increased Th1-type cellular immune response	[37]
<i>Allium cepa</i>	Juice	RH	Bulbs	Rat (Wistar albino)	1 ml	30 days	Orally	Tachyzoite	Kidneys Apoptosis in Toxoplasma group significantly increased in comparison to control group (P < 0.05)	[38]
<i>Zingiber officinale</i> (GE/F1)	Methanol	RH	Roots	Mice (BALB-c)	250 and 500 µg/ml	5 days	i.p	Tachyzoite	Results showed GE/F1 effectively inhibited the secretions of INF-γ and IL-8	[7]
<i>Thymus broussonetii</i> Boiss	Essential oil	Prugnialud (PRU)	Aerial parts	Mice (OF1)	20 µg	30 min, 3 days, and 6 days	Orally	Cyst	These essential oils of thyme have a blocking effect on the appearance of the cysts	[39]

Table 2 (continued)

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Fraxinus rhycho-phylla</i>	Oleuropein	RH	?	Mice (KP100 CD-1)	42, 125 and 300 mg/kg	4 days	Orally	Tachyzoite	The inhibition ratio of <i>T. gondii</i> in the peritoneal cavity was 55.4% compared to the negative control group after treatment with 300 mg/kg oleuropein	[21]
<i>Piper betle</i> L.	Ethanol	RH-GFP and PLK-GFP	Leaves	Mice (C57BL/6)	400 and 100 mg/kg	7 day	i.p	RH-GFP tachyzoites	Treatment with 400 mg/kg of the <i>P. betle</i> extract resulted in 100% mouse survival following infection with 100,000 tachyzoites	[22]
<i>Radix clematidis</i> <i>Fructus Ulmi</i> <i>Macrocarpae</i> <i>Atractylodes Anomala</i> <i>Glycyrrhizae</i>	Water	RH	Aerial parts	Mice (ICR)	300, 400, 500, and 600 mg/day	Single dosages	Orally	Tachyzoite	The survival time of Mice 500 mg Chinese herb group (9.8 ± 0.69) The livers of 500 mg Chinese herbs-treated mice displayed few changes compared to normal mice	[40]

Table 2 (continued)

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Allium Cepa</i>	Methanol	RH	Bulbs	Rats (Wistar)	1 ml	30 days	Orally	Tachyzoites	Sperm count was significantly increased in the experimental group that received 1 g/kg freshly prepared onion juice as compared with the control group	[41]
<i>Commiphora myrrha</i>	Water	RH	?	Mice (CBA/J)	100 or 200 mg	3 days	i.p	Tachyzoite	Myrrh treatment significantly enhanced protection and decreased the tachyzoites number in experimental animals (96.6 and 100%)	[42]

Table 2 (continued)

Family and botanical name	Preparation	Organism tested	Part of used	Type of animal	Concentration	Treatment durations	Routes of administration	Infection routes (cysts or tachyzoites)	Result	References
<i>Phoenix dactylifera</i> <i>Ziziphus spina-christi</i>	Methanol	RH	Seeds leaves	Female Balb/c mice	100 mg/kg	2 weeks	Orally	Tachyzoite	AgNPs (green synthesized via date palm and nabka) increased survival rate 2-AgNPs (green synthesized via date palm and nabka) decreased ALT and AST 3-AgNPs (green synthesized via date palm and nabka) decreased proinflammatory cytokines as TNF- α and IFN- γ	[43]

Sophora flavescens

In the traditional Chinese medicine, this plant is used as an antibacterial, antiviral and anti-tumor drug. The pharmacokinetics function of this plant is related to alkoids, such as oxymatrin (om) and matrine (ME), which have antioxidant, anti-tumor, and anti-diabetic properties [9, 54]. Previous studies indicated that the effect of *Sophora alkoid* on toxoplasmosis control, *in vitro* and *in vivo*, reduces tachyzoite in infected mice and also improves hepatic enzymes, such as ALT, AST, GSH and MDA [9].

Lycopodium clavatum

Lycopodium clavatum, known as Wolf Paw Clubmoss, is a dry moss belongs to Lycopodiaceae species. Geographically, it is widely distributed in the northern hemisphere and can also grow in non-fertile soils. In medicine, extracts of this plant are used in the treatment of acute diseases, such as children's autism as well as chronic diseases, such as Alzheimer's disease, and also in homeopathy [31]. The different effects of *L. clavatum* 200 dH on the treatment of clinical and parasitic aspects of mice infected with *T. gondii* (ME-49 strain) were evaluated. In this study, the mice of the G72 group were treated with *L. clavatum* 200dH for three consecutive days before infection with parasite strain. The lesions became worse in the genital area and the conversion rate of tachyzoites to bradyzoites and formation of brain cysts was higher than the G48 group that encountered parasite strain 48 h before the infection and were treated with *L. clavatum* 200 dH on days 2, 4 and 6 after infection. In the G48 group, the conversion rate of tachyzoites to bradyzoites was lower [31].

Sambucus nigra

Sambucus nigra has been used in the treatment of burns, wound injuries, rheumatism, hemorrhoids and common colds in traditional medicine. Moreover, the antiviral, antiprotozoal and antiprotozoal effects of this plant are also well known [15]. The antioxidant properties of this extract as well as its therapeutic effect are noted in the treatment of diabetes [55]. In past study, the methanolic extracts from the fruits and leaves of *Sambucus nigra* at the concentrations of 5 mg/ml and 10 mg/ml after 60 min and 120 min resulted in the excretion of 100% *T. gondii* tachyzoites. Besides, the concentrations of 25 mg/ml and 50 mg/ml of *S. nigra* fruit extract resulted in the death of 100% tachyzoates. So, the *S. nigra* fruit extract had a better parasite excretion effect than the leaf extract *in vitro* [15].

Berberis vulgaris

Berberis vulgaris belongs to the Berberidaceae family, which is known as barberry, and is widely grown in most parts of the world, especially in Iran. In traditional medicine, different parts of this plant like roots, leaves and fruits have been used in the treatment of skin, respiratory, digestive and cardiovascular diseases. Furthermore, in recent studies, the antimicrobial, anti-parasitic and anti-inflammatory effects of this plant have been proven [33, 56]. Mahmoudvand *et al.* evaluated the methanolic extract of *B. vulgaris* in the mice infected with *T. gondii*. In the mice that were treated with *B. vulgaris* at the concentrations of 1 and 2 g/kg, the time of death was postponed with 100% mortality rate on day 9. However, in other groups and in the control group, the mice died on day 5. Also, the mean number of tachyzoites in the treated group in doses 1 and 2 was 79×10^4 and 131×10^4 , respectively, while the number of tachyzoites in the control group was equal to 288×10^4 , where *B. vulgaris* could play a proactive role in toxoplasmosis [33].

Dictamnus dasycarpus

Dictamnus dasycarpus is used in traditional Chinese medicine for the treatment of various diseases, such as jaundice, cough, eczema, rubella, gall, some skin diseases and rheumatism [57]. *In vitro* study, the activity of *D. dasycarpus* ethanolic extract had anti-toxoplasmosis and cytotoxic activities, and the optimal and selective concentration of *D. dasycarpus* extract was equal to 7.52, which was so much higher than sulfadiazine at the concentration of 2.08. It also inhibited the proliferation of *T. gondii in vivo* [58].

Centaurea lydia* and *Phlomis nissolii

Centaurea lydia belongs to the Asteraceae family, which is widely found in the Mediterranean and Western Asia, having antioxidant, antimicrobial and anti-inflammatory properties. In the compounds, the structure sesquiterpene resulting from *Centaurea* specie is known as cytotoxic and antitumor. *P. nissolii* belongs to the Labiatae family enriched with flavonoid had anti-diabetic, anti-inflammatory and analgesic effects in the *in vivo* conditions, as well as anti-bacterial, anti-fungal, and anti-cancer effects in the *in vitro* conditions [59]. The study of Sigva *et al.*, the role of *C. lydia* and *P. nissolii* against *T. gondii* was investigated. *C. lydia* resulted in a 85% decrease in tachyzoites in this group of mice. It also increased survival time to 7 days compared to the control group and a group of mice treated to *P. nissolii*. So, the Lidia extract is an appropriate drug for the treatment of toxoplasmosis [59].

***Myristica fragrans* Houtt**

The seeds of Nutmeg, also known as *Myristica fragrans* Houtt, are used in the treatment of rheumatism, diarrhea, asthma, atherosclerosis and abdominal bloating, the anti-inflammatory effects of which have also been proven [16]. Pillai *et al.* [16] evaluated its anti-toxoplasmosis effect *in vitro* using the extract of essential oil derived from nutmeg (*M. fragrans* Houtt), where the essential oil of nutmeg with EC₅₀ (24.45 µg/ml) had anti-parasitic effects, and this anti-parasitic effect might be due to the compounds of myristicin and limonene in the nutmeg essential oil [16].

***Myrtus communis* and *Artemisia aucheri* Boiss**

Myrtus communis is a species of the Myrtaceae family that is used for treating urinary tract infections, digestive problems, vaginal discharge, bronchial congestion, sinusitis, dry cough as well as acne, ulcers, gum infections and hemorrhoids. *Artemisia aucheri* Boiss is a plant from the Asteraceae family that is used as soothing, antiseptic, antimicrobial and anti-parasitic agents in traditional medicine [17]. The results of study showed that the extract of these two plants had parasite excretion properties, but EC₅₀ and features of these two extracts were lower than pyrimethamine and their extract cannot be a good alternative for anti-*T. gondii* drugs [17].

Bunium persicum

Bunium persicum belongs to the Apiaceae family, which is widely grown in the southeast of Iran. The plant seeds are used in traditional medicine as antispasmodic, anti-flatulence, anti-epilepsy and breast milk increase. Other studies have also demonstrated that the essential oil of this drug is an anti-inflammatory, antioxidant, anti-parasitic and antibacterial agent [34, 60]. Tavakoli *et al.* (2015) found in the oral prescription of the essential oil of *B. persicum* by 0.05 and 0.1 ml/kg twice a day for 5 days, the mortality rate in the mice infected with *T. gondii* group was delayed by 8 and 9 days, but the control mice died on 5 days. Besides, the 0.1 ml/kg concentration of *B. persicum* significantly reduced the number of tachyzoites in this group compared to other groups [34].

Eurycoma longifolia

Eurycoma longifolia Jack from Simaroubaceae family is locally known as "Tongkat Ali" or "Pasakbumi", which is often used in traditional medicine for the treatment of nausea, swollen lymph nodes and fever. This plant is found in Asian countries, such as India, China, Thailand, Malaysia and the Philippines. In other studies, the cytotoxic effects of this plant against malaria have been proven [10]. Kavitha *et*

al. referred to the role of crude extracts and fractions (TAF 273, TAF 355, TAF 191 and TAF 401) derived from *E. longifolia* against *T. gondii*. They showed the fractions TAF 355 (EC₅₀=0.369 µg/ml) and TAF 401 (EC₅₀=0.882 µg/ml) were used as a new product against *T. gondii*. The treatment of parasite-infected cells with fractions reduced intracellular tachyzoites and prevented the proliferation and destruction of the infected cells [10].

Astragalus membranaceus

This herb is one of the most popular drugs in China with a history of about 2000 years ago. Various components of this plant, especially the polysaccharide section, have a medicinal function, but the molecular mechanism of this effect has not yet been identified [61, 62]. As shown, this plant increases specific immune responses [63]. In a study, using the extract of *Astragalus membranaceus* (0.075 mg/g) combined with atovaquone (100 mg/kg/day) in the mice infected with parasites through oral administration reduced the number of trophozoites ($p < 0.05$). Furthermore, the IL-12 cytokine level was significantly reduced compared to the control group ($p < 0.05$), and IFN- γ cytokine level in a group treated with Ato + Astra was increased compared to the control one [64]. Using the aqueous extract of *A. membranaceus* and baicalensis, GEORGI Scutellaria, the immune response induced in the ICR mice infected with *T. gondii* was reduced to UV radiation. The ICR mice treated with AmE and SbE challenged with RH strain tachyzoites had significantly longer survival time, decreased parasitic load and higher Th1 cellular immune response. Besides, the *T. gondii* reduced to UV radiation injected into the ICR mice with AmE and SbE reduced the number of tachyzoites and increased toxoplasma specific IgG, especially IgG 2a [37].

Achillea millefolium* and *Hypericum perforatum

Hypericum perforatum from Hypericaceae family is one of the medicinal herbs, the extract of which contains more than 25 active compounds, where hyperbilirubin, hyperurine and flavonoid compounds have been reported as major metabolites, with various anti-depressant, antimicrobial, antioxidant and immune system stimulating effects. It also heals lesions and burns, and reduces the effect of bite caused by insects [18, 65]. *Achillea millefolium*, known as Yarrow, belongs to the Asteraceae or Compositae family. The active ingredient of this plant is an essential oil, composed of kemazolen, cineol, camphor and limonene; the flavonoids with urinary tract disinfection, analgesic, antispasmodic, anti-hemorrhage and infection, anti-flatulent and anti-inflammatory effects, as well as increased urine volume and renal stone therapy [18, 66]. Nozari *et al.* studied the effect of *A. millefolium* and *H. Perforatum*

on *T. gondii* tachyzoites. The concentrations of 100 mg/ml of both herb extracts were killed all tachyzoites, while in the concentration of 10 mg/ml, the tachyzoites mortality was 4.53–5.31% [18].

Satureja khuzestanica

Satureja khuzestanica Jamzad is a medicinal plant that usually grows in western Iran. Studies have shown the analgesic effects in this extract, as well as anti-inflammatory, antioxidant, antimicrobial and anti-diabetic properties [67, 68]. The *S. khuzestanica* essential oil used in the mice infected with *T. gondii* tachyzoites at the dose of 0.2 and 0.3 ml/kg/day for 2 weeks in the control and treated groups that led to delays in the death from 5 to 9 days, respectively. Moreover, it was reduced the number of tachyzoites in the treated [35].

Nigella sativa

Nigella sativa from the Ranunculaceae family, known as black seed, has anti-inflammatory, anti-parasitic, antimicrobial and disinfectant effects that are properly identified [69]. The anti-parasitic and anticoccidial effects of this plant are also mentioned. This plant also stimulates the immune system, increases the production of interleukin and strengthens the cellular immune response [36]. The study of Rayan *et al.* showed that the role of black seed oil (BSO) of *N. sativa* against infection of the Swiss albino mice with strain ME49 (cyst-shaped) of the *T. gondii* was mentioned. This research showed that BSO increased the survival rate of infected mice and reduced the number of brain cysts compared to the control mice. A mild meningitis and encephalitis could be seen in the Swiss albino mice treated with BSO [36].

Ginkgo biloba

Ginkgo biloba sarcotesta has been used for thousands of years in traditional Chinese medicine. Ginkcolic acid (GA)-derived *G. biloba* sarcotesta had an antimicrobial activity and anti-tumor function [19]. In other research, the effect of fresh onion juice was noted as a strong antioxidant, which reduced the renal damage in the mice infected with *T. gondii* [38]. Chen *et al.* pointed out the anti-toxoplasmosis activity of ginkgolic acids (GAs) derived from ginkgo biloba *in vitro*. human foreskin fibroblast (HFF) cells infected with toxoplasma were exposed to (GAs) for 48 h, no trophozoite was seen in the cells. Besides, by evaluating the electron microscopy, the destruction of HFF cells was not observed due to parasite proliferation and, thus, changes in cell morphology [19].

***Thymus broussonetii* Boiss**

Thymus is one of the variety of herbs used in cooking and medicines, which is known as an antimicrobial, antispasmodic, anti-inflammatory and antioxidant medicine. *Thymus* genus has several species, such as *T. broussonetii*, which is used in the treatment of diarrhea and fever [39]. Previous data have indicated that, use of the essential oil of *T. broussonetii* Boiss prevented the development of cystic shapes of *T. gondii* in the mice. In this study, in the rat groups that were initially infected with cystic shapes and, then, received *T. hymus broussonetii*, the cystic shapes in the brain were not seen in the days after the infection of the mice (OF1), which could be due to the inhibitory role of essential oils in the destruction of prescriptive cysts and the release of bradyzoites in the intestines of the mice [39].

Artemisia annua

Artemisin, which is derived from *Artemisia annua*, is one of the local herbs used in traditional Chinese medicine. It is often used without toxicity properties to improve *Plasmodium* infections [70, 71]. The anti-parasitic properties of this herb against the *Trypanosoma cruzi* have been proven in the labs [72]. It has also been shown that the *Schistosoma* parasites have good effects against the immature worms. They also have *Schistosoma* preventing effects [73]. In another study, the effects of artesunate excretion were evaluated on the RH and ME49 strains of *T. gondii* *in vitro* and *in vivo*. The low survival rate for the tachyzoites of the RH strain at the concentration of 0.1 µg/ml was 16.67% for 48 h, while it was 41.67% for 24 h, so a significant difference was found. The apoptosis process of tachyzoites, such as wrinkling, cytoplasm vacuolization and membrane changes in tachyzoites, was seen. In *in vivo*, the artesunate effect on ME49 strains significantly reduced the number of brain cysts in the Swiss albino mice [8].

Piper betle

Piperaceae plants almost make up 1000 species of plants found in the tropical regions of India, Southeast Asia and Africa. Forty species of these plants have been identified in Thailand. They are used as active ingredients in traditional Thai medicine for stress reduction, digestion, nutrient uptake and general health balance. Pharmacologically, the properties of Piperaceae plants are known as antibacterial, antioxidant, lipid protector and anticancer. Additionally, these studies have represented that this plant has anti-leishmaniasis and antimalarial activity [22]. In a study, the *P. betle* extract was the most effective extract in inhibiting the proliferation of *T. gondii* in human foreskin fibroblast (HFF) cells. After the treatment of mice with 400 mg/kg of the *P. betle* extract

for 7 days and challenge of mice with 1000 tachyzoites from PLK strain of the *T. gondii*, result in 100% survival rate was increased [22].

Radix clematidis

Radix is used in traditional Chinese medicine as an anti-cancer, anti-inflammatory and antimicrobial drug [40]. In a preceding research, a plant compound consisting of *R. clematidis* (10 g), *Fructus Ulmi Macrocarpae* (8 g), *Atractylodes* (15 g), *A. anomala* (10 g) and *Glycyrrhizae* (6 g) with the injection dose of 500 mg to the mice led to an increased survival time and slow proliferation of *T. gondii* tachyzoites in the treated mice [40].

***Tinospora crispa* Miers**

Tinospora crispa Miers (from Menispermaceae family) is used for the treatment of jaundice, rheumatism, fever, filariasis and malaria, and is also an anti-inflammatory and antispasmodic agent [23]. Lee *et al.* investigated *T. crispa* stem extract with $EC_{71} = 7.71 \pm 1.56$ µg/ml had an anti-toxoplasmosis activity. Also, fractions 4 and 5 of *T. crispa* with $EC_{50} = 6.71 \pm 1.95$ and $EC_{50} = 6.06 \pm 2.51$, respectively, had the best inhibitory concentration for the proliferation of *T. gondii* [23]. Sharif *et al.* assessed the effect of *T. crispa* (EETC) on the host cell invasion and intracellular replication of *T. gondii*, that the EETC had excellent anti-toxoplasma activities with $IC_{50} = 6.31$ µg/ml in comparison with clindamycin ($IC_{50} = 8.33$ µg/ml) and Veratrine alkaloid ($IC_{50} = 14.25$ µg/ml) [24].

Tectona grandis* and *Vernonia amygdalina

As a shrub that grows across Africa, it is used for cooking or drinking in some parts of Africa. This plant has a laxative effect, as well as anti-worm and anti-malarial effects [74]. In a study, the ethanolic and hydroethanolic extracts of the skin and leaves of *Tectona grandis* and the ethanolic extract of *Vernonia amygdalina* had inhibitory effects on toxoplasmosis. Also, the ethanolic extract of *V. amygdalina* increased the production of TNF-α and NO, which is one of the pro-inflammatory factors for the inhibition of *T. gondii* *in vivo* [25]. The fresh onion juice in the rats infected with *T. gondii* reduced the harmful effects of the disease and increased the fertility and sperm quality, as well as anti-toxoplasmosis IgG and IgM responses than other groups [41].

Jatropha curcas

Jatropha curcas is a species of leafy tree belonging to the family Euphorbiaceae. This plant contains drought-resistant oil seeds found in tropical and subtropical areas.

Different parts of this plant are used in medical treatment, e.g. its skin is used in the treatment of lesions and hemostasis, the root is used in leukemia and diuretic, and leaves are used in the treatment of skin diseases. *J. curcas* also has anticoagulant effects. Many studies have shown that the leaves of this plant have an excretion property for nematodes, fungi, larvae and insects [26, 75]. Soares *et al.* demonstrated that the role of a protein extract and a fraction enriched with inhibitory proteases derived from *J. curcas* seed cake were noted in the proliferation of *T. gondii* tachyzoites. Accordingly, different concentrations of JCSC during 24 h *in vitro* had an anti-parasitic activity and reduced the number of tachyzoites in the contaminated cells [26].

Cinnamomum zeylanicum

The leaves of these plants are known as an antioxidant, anti-carcinogenic, anti-hyperglycemia, anti-inflammatory, antimicrobial, anti-parasitic and antiviral drugs [27, 76]. Preceding study showed using the aqueous extract of neem and cinnamon during 24 h *in vitro*, the percentage of infected cells and the number of intracellular *T. gondii* tachyzoites were significantly reduced. The aqueous extract of neem at the concentration of 5 mg/ml reduced the infection by 85%, while the aqueous extract of cinnamon at the concentration of 1 mg/ml reduced 70% of infections in parasite-infected cells without causing any toxicity to the cells [27].

Scutellaria baicalensis

These two plants are derived from Huang-Qi that have been used since centuries ago as a traditional medicine in China for the treatment of infectious and inflammatory diseases. The pharmacological and clinical studies show that *A. membranaceus* is capable of inducing and stimulating the immune system, both *in vivo* and *in vitro*, and is widely used to improve the natural defense mechanisms of the body and to cure different immune disorders. Moreover, the plant is widely used to treat cardiovascular diseases in China. Also, *S. baicalensis* has been using an anti-bacterial and antiviral drug for the treatment of respiratory diseases, diarrhea, hepatitis and jaundice [28, 77]. The anti-inflammatory effects of *S. baicalensis* are also mentioned [78]. Yang *et al.* [28] proved that the effect of *A. membranaceus* and *S. baicalensis* for inhibiting the proliferation of *T. gondii* tachyzoites the tachyzoites were marked with green protein fluorescent (GFP) and the cells containing RH-GFP tachyzoites treated with SbE and AmE were significantly reduced compared to the untreated or control groups. [78].

Phoenix dactylifera and Ziziphus spina-christi

The genus *Ziziphus* belongs to the Rhamnaceae family which consists of about 100 species that distributed in the tropical and subtropical areas of the world [79, 80]. The genus *Ziziphus* is a multipurpose fruit tree and famous for its medicinal properties as a liver protective factor, hypoglycemic agent, hypotensive, antioxidant, antimicrobial, anti-inflammatory, and as an immune system stimulating [81]. *Phoenix dactylifera* L. (Date palm) is considered as one of the oldest and ancient crops which is native to the tropical or subtropical regions of southern Asia or Africa [82]. Recent research showed that Date fruits may have numerous potential health benefits due to highly nourishing. Moreover, the protective effects of date fruits against chronic diseases are recognized to bioactive non-nutrients called phytochemicals [83, 84]. Alajmi *et al.* evaluated the anti-toxoplasmic effect of silver nanoparticles green synthesized with Date palm and *Ziziphus spinachristi* extracts. Based on the improvement of their results in the histological features indicated by slight infiltration and fibrosis, minimal pleomorphism and less hepatocyte and degeneration were shown. Also, nanoparticle treatment made a reduction in immunoreactivity of TGF- β and NF- κ B in hepatic tissues [43].

Limitations

One of the problems that we confront in drug therapy is the lack of study for identifying the effective compound and most of these drugs have not been evaluated *in vivo* and their efficacy against tissue cysts is also unclear.

Conclusion

One of the potential applications of medicinal plant extracts with anti-toxoplasma effect is to prevent the congenital toxoplasmosis and re-activation of *Toxoplasma* in immunocompromised patients. In some cases, using the plant extracts, not only the proliferation rate of the parasite is reduced, but also the life span of laboratory animal is increased.

Currently, synthetic drugs are used for this purpose and the side-effects are the most important inhibiting factor for them. It seems that herbal medicine against *T. gondii* can be a good alternative to synthetic drugs in terms of preventing toxoplasmosis in these individuals. All the drugs evaluated in this study can reduce the clinical signs and symptoms, and can even eliminate these symptoms, while in many cases, they cannot remove the parasite or cure the infected host. However, further studies seem necessary to identify the mechanism of effective compounds and their extracts. This requires extensive analysis of compounds capable of

parasite excretion with minimum complications and toxicity for human beings.

Compliance with Ethical Standards

Conflict of interest The authors declare no conflict of interest in this study.

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